

VAN / SURNAME: MEMO

VOORLETTERS / INITIALS: _____ REG.No.: _____

HANDTEKENING / SIGNATURE: _____

KURSUS / COURSE: FSK 176

UNIVERSITEIT VAN PRETORIA
THE UNIVERSITY OF PRETORIA

FSK 176
EKSAMEN / EXAMINATION
11 NOVEMBER 2011

INTERNE EKSAMINATOR / INTERNAL EXAMINER: PROF H W ALBERTS
EKSTERNE EKSAMINATOR / EXTERNAL EXAMINER: DR J NEL

Totale Punte / Total Marks: 125

Tyd / Time: 3 ure / hours

- Beantwoord alle vrae. / Answer all questions.
- Programmeerbare sakrekenaars mag NIE gebruik word NIE
- Programmable pocket calculators may NOT be used.
- Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.
All test and exam regulations of the University of Pretoria are applicable.

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FSK 176: EKSAMEN / EXAMINATION
11 NOVEMBER 2011

VAN EN VOORLETTERS: _____
SURNAME AND INITIALS:

STUDENTE NR. _____
STUDENT NUMBER:

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VRAAG/QUESTION 1

- (a) Die uitwyking y van 'n golf, wat met 'n snelheid v in die positiewe x -as rigting beweeg, word op tyd $t = 0$ gegee deur:
The deviation y of a wave travelling in the positive x -direction with velocity v , is given at $t = 0$ as:

$$y(x, 0) = A \cos\left(\frac{2\pi x}{\lambda}\right)$$

- (i) Toon aan dat op enige tyd t die golffunksie geskryf kan word as:
Show that at any instant this wave function can be written as:

$$y(x, t) = A \cos(kx - \omega t)$$

$$\begin{aligned} y(x, t) &= A \cos\left[\frac{2\pi}{\lambda}(x - vt)\right] \\ &= A \cos\left[\frac{2\pi x}{\lambda} - \frac{2\pi}{\lambda}(vt)\right] \\ &= A \cos\left[kx - \frac{2\pi}{\lambda}(\lambda f)t\right] \\ &= A \cos[kx - 2\pi f t] \\ &= A \cos[kx - \omega t] \end{aligned} \quad (3)$$

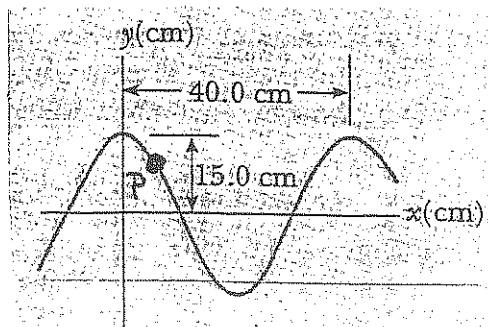
- (ii) Skryf uitdrukkings neer vir die golfgetal k en die hoeksnelheid ω en gee hulle eenhede.

Give expressions for the wave number k and the angular velocity ω . Also give their units.

$$\begin{aligned} k &= \frac{2\pi}{\lambda} \quad \checkmark \quad \text{rad/m} \quad \checkmark \\ \omega &= \frac{2\pi}{T} \quad \checkmark \quad \text{rad/s} \quad \checkmark \end{aligned} \quad (4)$$

- (b) 'n Golf $y = A \sin(kx - \omega t + \delta)$ beweeg met 'n snelheid v in die positiewe x -as rigting. Die amplitude van die golf is 15 cm, sy golflengte is 40 cm en sy frekwensie is 8 Hz. Met $t = 0$ en $x = 0$ is die vertikale verplasing $y = 15$ cm.

A wave $y = A \sin(kx - \omega t + \delta)$ is travelling in the positive x -direction with a velocity v . It has an amplitude of 15 cm, a wavelength of 40 cm and a frequency of 8 Hz. The vertical displacement at $t = 0$ and $x = 0$ is $y = 15$ cm.



- (i) Bepaal die golfgetal
- k
- en gee sy eenhede.

Determine the wave number k and give its units.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{40} = 0,157 \text{ rad/cm}$$

(2)

- (ii) Bepaal die hoeksnelheid
- ω
- en gee sy eenhede.

Determine the angular velocity ω and give its units.

$$\omega = 2\pi f = 2\pi(8) = 50,3 \text{ rad/s}$$

(2)

- (iii) Bepaal die periode
- T
- .

Determine the period T .

$$T = \frac{1}{f} = \frac{1}{8} = 0,125 \text{ s}$$

(2)

- (iv) Bepaal die spoed
- v
- van die golf.

Determine the speed v of the wave.

$$v = \lambda f = 40(8) = 320 \text{ cm/s}$$

(2)

- (v) Bepaal die fasekonstante
- δ
- , waar
- $x = 0$
- op
- $t = 0$

Determine the phase constant δ , where $x = 0$ at $t = 0$.

$$y = A \sin(kx - \omega t + \delta)$$

$$x=0, t=0 : y = A \sin \delta$$

$$15 = 15 \sin \delta$$

$$\therefore \sin \delta = 1$$

$$\delta = \pi/2$$

(3)

- (vi) Met die waardes soos hierbo bereken, herskryf nou die golf funksie

 $y = A \sin(kx - \omega t + \delta)$ sodat die fasehoek δ nie daarin voorkom nie.

With the values as calculated above, rewrite the wave function

 $y = A \sin(kx - \omega t + \delta)$ but in such a way that δ does not appear in the equation.

$$y = A \sin(kx - \omega t + \delta)$$

$$= 15 \sin(0,157x - 50,3t + \pi/2)$$

$$= 15 \cos(0,157x - 50,3t)$$

(3)

- (vii) Bereken die waarde van
- y
- (in cm) by
- $x = 3$
- cm en
- $t = 2$
- s.

Determine the value of y (in cm) at $x = 3$ cm and $t = 2$ s.

$$y = 15 \cos[0,157(3) - 50,3(2)]$$

$$= 13,8 \text{ cm}$$

(2)

- (viii) Toon met al die nodige stappe aan dat die maksimum snelheid van punt P in die
- y
- as rigting (by 'n vaste waarde van
- x
-) 754,5 cm/s sal wees.

With all the necessary steps, show that the maximum velocity of point P in the y direction (at a fixed value of x) is 754,5 cm/s.

$$y = A \cos(kx - \omega t)$$

$$v_y = \frac{dy}{dt} \bigg|_{x=c} = \omega A \sin(kx - \omega t)$$

$$\therefore v_{y \max} = \omega A$$

$$= 50,3 (15)$$

$$= 754,5 \text{ cm/s} \Rightarrow$$

(2)

[25]

VRAAG / QUESTION 2

- (a) 'n Voorwerp beweeg in 'n reguit lyn met 'n konstante versnelling a.
An object is moving in a straight line and is experiencing a constant acceleration a.

- (i) Toon deur middel van integrasie aan dat (vir $t_0 \neq 0$) is $v = v_0 + a(t - t_0)$
With integration, show that (with $t_0 \neq 0$) $v = v_0 + a(t - t_0)$

$$a = \frac{dv}{dt}$$

$$\int_{v_0}^v dv = \int_{t_0}^t a dt$$

$$[v]_{v_0}^v = a \int_{t_0}^t dt$$

$$v - v_0 = a[t - t_0]$$

$$\therefore v = v_0 + a(t - t_0)$$

(4)

- (ii) Deur middel van integrasie, toon ook aan dat
With integration, also show that

$$(x - x_0) = v_0(t - t_0) + \frac{1}{2}a(t - t_0)^2$$

$$v = \frac{dx}{dt} \quad dx = v dt$$

$$\int_{x_0}^x dx = \int_{t_0}^t v dt$$

$$= \int_{t_0}^t (v_0 + at - at_0) dt$$

$$[x]_{x_0}^x = [v_0 t + \frac{1}{2}at^2 - at_0 t]_{t_0}^t$$

$$x - x_0 = (v_0 t + \frac{1}{2}at^2 - at_0 t) - (v_0 t_0 + \frac{1}{2}at_0^2 - at_0 t_0)$$

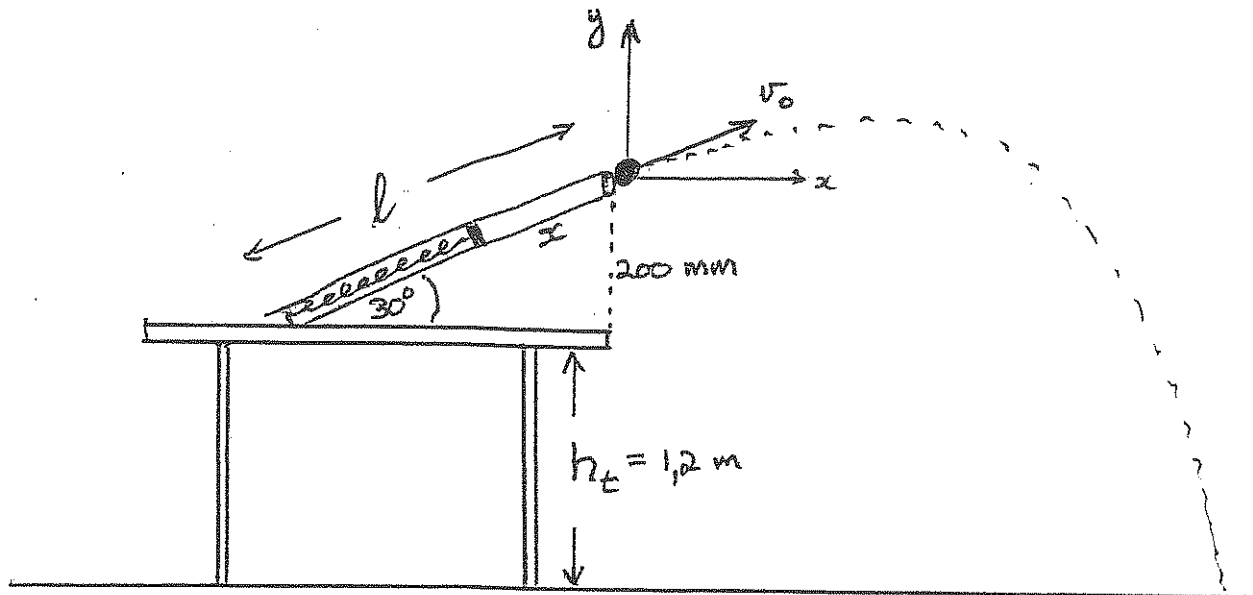
$$= v_0(t - t_0) + \frac{1}{2}at^2 - at_0 t + \frac{1}{2}at_0^2$$

$$= v_0(t - t_0) + \frac{1}{2}a\{t^2 - 2tt_0 + t_0^2\}$$

$$x - x_0 = v_0(t - t_0) + \frac{1}{2}a(t - t_0)^2$$

(5)

- (b) 'n Albaster (massa $m = 300$ gram) word geskiet met 'n geweer wat 'n veer bevat met weerkonstante $k = 155$ Nm. Die geweer maak 'n hoek $\theta = 30^\circ$ met die tafel (Sien skets).
A marble (mass $m = 300$ gram) is fired from a spring loaded gun (spring constant $k = 155$ Nm) that is mounted at an angle $\theta = 30^\circ$ on a table, as shown in the figure.



- (i) Bereken die lengte ℓ van die geweer (in meter).

Determine the length ℓ of the gun (in meter).

$$\sin 30^\circ = \frac{200}{\ell}$$

$$\ell = \frac{200}{\sin 30^\circ} = 400 \text{ mm} = 0,4 \text{ m} \Rightarrow$$

(1)

- (ii) Die veer word nou sodanig saamgedruk dat sy lengte $\frac{2}{3}$ is van die ontspanne toestand, waar dit dieselfde lengte gehad het as die geweer se loop. Bepaal x (in m).
In the relaxed position the spring has the same length as the barrel. When the spring is compressed its length is $\frac{2}{3}$ that of the barrel. Determine x (in m).

$$x = \frac{1}{3} \ell$$

$$= \frac{1}{3} (0,4) = 0,133 \text{ m} \Rightarrow$$

(1)

- (iii) Bereken die energie E_v wat in die saamgedrukte veer gestoor is.

Determine the energy E_v stored in the compressed spring.

$$E_v = \frac{1}{2} k x^2$$

$$= \frac{1}{2} (155) (0,133)^2$$

$$= 1,37 \text{ J} \Rightarrow$$

(2)

- (iv) Wanneer die veer losgelaat word en deur afstand x beweeg, toon aan dat die verandering van die potensiële energie van die bal $E_p = 0,2 \text{ J}$.
When the spring is released and has moved through the distance x , show that the change in the potential energy of the ball is $E_p = 0,2 \text{ J}$.

$$\begin{aligned} E_p &= mgh \checkmark \\ &= mg(x \sin 30^\circ) \checkmark \\ &= 0,3(9,8)(0,133 \sin 30^\circ) \\ &= 0,2 \text{ J} \Rightarrow \end{aligned}$$

- (v) Toon volledig aan dat die snelheid van die bal, wanneer dit die loop verlaat, is $v_0 = 2,8 \text{ m/s}$.
Show, with the necessary steps, that the velocity of the ball when it leaves the barrel, is $v_0 = 2,8 \text{ m/s}$.

$$\begin{aligned} \frac{1}{2} kx^2 \checkmark &= mgh \checkmark + \frac{1}{2} m v_0^2 \checkmark \\ 1,37 &= 0,2 + \frac{1}{2} (0,3) v_0^2 \\ v_0^2 &= \frac{2(1,37 - 0,2)}{0,3} \\ v &= 2,8 \text{ m/s} \Rightarrow \end{aligned}$$

- (vi) Met die oorsprong van die (x, y) -assesselsel by die uitgang van die loop (sien figuur), bereken die maksimum hoogte wat die bal bokant die uitgang kan bereik.
With the origin of the (x, y) axis at the exit point of the barrel (see figure), calculate the maximum height of the ball above the exit point.

$$\begin{aligned} v_y &= v_0^2 + 2as \checkmark \\ 0 &= (v_0 \sin 30^\circ)^2 - 2(9,8)s \\ s &= \frac{(2,8 \sin 30^\circ)^2}{2(9,8)} = 0,1 \text{ m} \Rightarrow \end{aligned}$$

- (vii) Met $y - y_0 = v_0 t + \frac{1}{2} at^2$, bereken die tyd wat die bal sal neem voordat dit die vloer tref.
With $y - y_0 = v_0 t + \frac{1}{2} at^2$, determine the time for the ball to reach the floor.

$$\begin{aligned} -1,4 - 0 &= (2,8 \sin 30^\circ)t - \frac{1}{2}(9,8)t^2 \\ 4,9 t^2 - 2,8 \sin 30^\circ t - 1,4 &= 0 \\ 4,9 t^2 - 1,4 t - 1,4 &= 0 \\ t &= \frac{-(-1,4) \pm \sqrt{(1,4)^2 - 4(4,9)(-1,4)}}{2(4,9)} \\ &= 0,69 \text{ s} \Rightarrow \end{aligned}$$

- (viii) Indien die afstand vanaf die geweer na die oorkantste muur 2 m is, bereken of die bal die muur of die vloer sal tref. Op tydstip $t_0 = 0$ is die bal by $x_0 = 0$.

With a distance of 2 m to the opposite wall, calculate if the ball will hit the wall or the floor. On time $t_0 = 0$ the ball is at $x_0 = 0$.

$$x - x_0 = v_0 t + \frac{1}{2} a t^2$$

$$x - 0 = (2,8 \cos 30^\circ) (0,69) + 0$$

$$x = 1,67 \text{ m}$$

Hit floor / tref vloer.

(4)
[28]

VRAAG / QUESTION 3

- (a) Definieer 'n hoek θ , gemeet in radiale.

Define an angle θ , measured in radians.

$$\theta = \frac{s/r}{(\text{radius / string})}$$

(1)

- (b) Definieer hoeksnelheid ω en gee sy eenhede.

Define angular velocity ω and give its units.

$$\omega = \frac{d\theta}{dt}$$

(2)

- (c) Herlei 'n uitdrukking wat die verband gee tussen lineêre snelheid v en hoeksnelheid ω .

Deduce an equation which gives the relation between linear velocity v and angular velocity ω .

$$\omega = \frac{d\theta}{dt}$$

$$= \frac{d}{dt} \left(\frac{s}{r} \right)$$

$$= \frac{1}{r} \left(\frac{ds}{dt} \right)$$

$$= \frac{1}{r} v$$

$$v = \omega r$$

(3)

- (d) 'n Voorwerp ondergaan 'n konstante hoekversnelling α . Definieer hoekversnelling α .

An object has a constant angular acceleration α . Define angular acceleration α .

$$\alpha = \frac{d\omega}{dt}$$

(1)

- (e) Toon deur middel van integrasie aan dat:
Show with integration that:

$$\omega = \omega_0 + \alpha t$$

$$\begin{aligned} d\omega &= \alpha dt \\ \int_{\omega_0}^{\omega} d\omega &= \alpha \int_{t=0}^t dt \quad \checkmark \\ \omega - \omega_0 \quad \checkmark &= \alpha(t) \\ \omega &= \omega_0 + \alpha t \Rightarrow \end{aligned} \quad (2)$$

- (f) 'n Liggaam, met massa m , is opgedeel in massadeeltjies Δm_i . Die liggaam roteer met hoeksnelheid ω . Die lineêre snelheid van die deeltjie Δm_i is v_i en dié deeltjie is op afstand R_i vanaf die rotasie-as. Toon aan dat die rotasie kinetiese energie van die liggaam gegee word deur $K = \frac{1}{2} I \omega^2$, en gee ook 'n uitdrukking vir die traagheids-moment I .

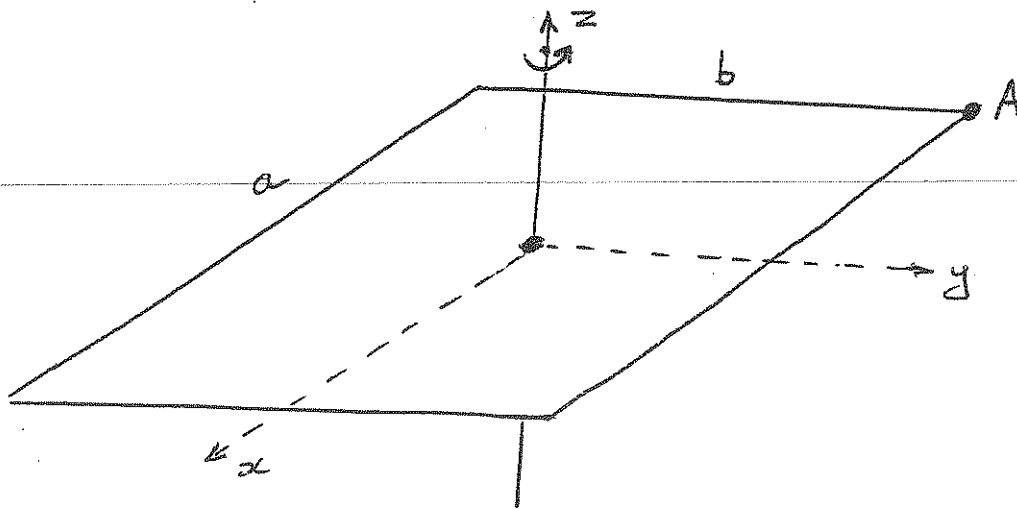
An object with mass m is divided in small mass particles Δm_i . The object is rotating with angular velocity ω . The linear velocity of particle Δm_i , which is at distance R_i from the axis of rotation, is v_i . Show that the rotational kinetic energy of the object is given by $K = \frac{1}{2} I \omega^2$, and also give an expression for the moment of inertia I .

$$\begin{aligned} K &= \sum K_i = \frac{1}{2} \sum \Delta m_i v_i^2 \quad \checkmark \\ &= \frac{1}{2} \sum \Delta m_i (\omega R_i)^2 \quad \checkmark \\ &= \frac{1}{2} \sum (\Delta m_i R_i^2) \omega^2 \quad \checkmark \\ &= \frac{1}{2} I \omega^2 \\ \text{with } I &= \sum \Delta m_i R_i^2 = m R^2 \quad \checkmark \end{aligned} \quad (4)$$

- (g) Die traagheidsmoment om 'n as deur die swaartepunt (CM) van 'n liggaam met massa M is I_{CM} . Skryf 'n uitdrukking neer vir die traagheidsmoment I van die liggaam om enige as wat parallel is aan die as deur die swaartepunt, maar op 'n afstand d daarvan.
The moment of inertia about an axis through the centre of mass (CM) of an object with mass M is I_{CM} . Give an expression for the moment of inertia I about any axis which is parallel to the one through the CM, but a distance d from it.

$$I = I_{CM} + M d^2 \quad (1)$$

- (h) 'n Plat plaat met sylengtes a en b roteer om 'n as deur sy swaartepunt. Toon aan dat die traagheidsmoment om 'n as deur die swaartepunt gegee word deur $I_{CM} = \frac{1}{12} M(a^2 + b^2)$.
A flat plate with sides a and b rotates about an axis through its center of mass. Show that the moment of inertia through the centre of mass is given by $I_{CM} = \frac{1}{12} M(a^2 + b^2)$.



$$\sigma = \frac{M}{ab} \quad \checkmark = \frac{dm}{dx dy} \quad \checkmark$$

$$\therefore dm = \frac{M}{ab} dx dy$$

$$\text{and } r^2 = x^2 + y^2 \quad \checkmark$$

$$I = \int r^2 dm \quad \checkmark$$

$$= \iint (x^2 + y^2) \left\{ \frac{M}{ab} dx dy \right\}$$

$$= \frac{M}{ab} \left\{ \int_{-a/2}^{a/2} x^2 dx \int_{-b/2}^{b/2} dy + \int_{-a/2}^{a/2} dx \int_{-b/2}^{b/2} y^2 dy \right\}$$

$$= \frac{M}{ab} \left\{ \left[\frac{1}{3} x^3 \right]_{-a/2}^{a/2} \left[y \right]_{-b/2}^{b/2} + \left[x \right]_{-a/2}^{a/2} \left[\frac{1}{3} y^3 \right]_{-b/2}^{b/2} \right\}$$

$$= \frac{M}{ab} \left\{ \left[\frac{1}{3} \left(\frac{a}{2} \right)^3 - \frac{1}{3} \left(-\frac{a}{2} \right)^3 \right] \left[\frac{b}{2} - \left(-\frac{b}{2} \right) \right] + \left[\frac{a}{2} - \left(-\frac{a}{2} \right) \right] \left[\frac{1}{3} \left(\frac{b}{2} \right)^3 - \frac{1}{3} \left(-\frac{b}{2} \right)^3 \right] \right\}$$

$$= \frac{M}{ab} \left\{ \frac{1}{12} a^3 b + a \frac{1}{12} b^3 \right\} \quad \checkmark$$

$$= \frac{M}{12} (a^2 + b^2) \Rightarrow \quad (7)$$

- (i) Bepaal die traagheidsmoment om 'n as wat deur A gaan en parallel is aan die z-as.

Determine the moment of inertia about an axis through A and which is parallel to the z-axis.

$$I = I_{cm} + Md^2 \quad \checkmark$$

$$= \frac{1}{12} M (a^2 + b^2) + M \left[\left(\frac{a}{2} \right)^2 + \left(\frac{b}{2} \right)^2 \right] \quad \checkmark$$

$$= \frac{1}{12} M (a^2 + b^2) + M \left(\frac{a^2}{4} + \frac{b^2}{4} \right)$$

$$= \left(\frac{1}{12} + \frac{1}{4} \right) M (a^2 + b^2)$$

$$= \frac{1}{3} M (a^2 + b^2) \Rightarrow \quad \checkmark \quad (4)$$

VRAAG / QUESTION 4

- (a) Wat is die verskil tussen die volgende tipes golwe?
What is the difference between the following types of waves?

* Longitudinale golwe / Longitudinal waves.

Displacement (verplasing) // Direction of propagation (roegplantings) (1) ^{↑ regting}

* Transversale golwe / Transverse waves.

" " " " " " (1)

- (b) Wat is die verband tussen die periode T van 'n golf en sy frekwensie f ?
Give the relation between the period T and the frequency f of a wave.

$$f = \frac{1}{T}$$

- (c) Gee 'n uitdrukking vir die spoed v van 'n golf in terme van sy golflengte λ .
Give the relation between the speed v of a wave and its wavelength λ .

$$v = \lambda f \quad \left(\text{or } v = \frac{\lambda}{T} \right)$$

- (d) 'n Bron met frekwensie f_0 beweeg met 'n spoed v_s na 'n stilstaande waarnemer. Indien die spoed van klank v is, bepaal:

A source with frequency f_0 is moving with speed v_s towards a receiver at rest. If the speed of sound is v , determine:

- (i) 'n Uitdrukking vir die nuwe golflengte λ' . / *An expression for the the new wavelength λ' .*

$$\begin{aligned} \lambda' &= \lambda_0 - \frac{v_s}{f_0} \checkmark \\ &= \frac{v}{f_0} - \frac{v_s}{f_0} \\ \therefore \lambda' &= \frac{v - v_s}{f_0} \Rightarrow \checkmark \end{aligned}$$

- (ii) Bepaal die frekwensie wat die waarnemer hoor.
Determine the frequency observed by the receiver.

$$\begin{aligned} f' &= \frac{v}{\lambda'} \checkmark \\ &= \frac{v}{(v - v_s)/f_0} \\ &= f_0 \frac{v}{v - v_s} \checkmark \\ &= f_0 \left(\frac{1}{1 - v_s/v} \right) \end{aligned}$$

- (e) 'n Waarnemer beweeg met spoed v_r na 'n stilstandende bron met frekwensie f_0' . Indien die spoed van klank v is, toon aan dat die nuwe frekwensie $f_1' = f_0' \left(1 + \frac{|v_r|}{v}\right)$ sal wees.

A receiver moves with velocity v_r towards a stationary source with frequency f_0' . If the speed of sound is v , show that the new frequency will be $f_1' = f_0' \left(1 + \frac{|v_r|}{v}\right)$.

New speed $v' = v + |v_r|$

$$\begin{aligned} \therefore f_1' &= \frac{v'}{\lambda} = \frac{v + |v_r|}{\lambda} \\ &= \frac{v}{\lambda} + \frac{|v_r|}{\lambda} \\ &= f_0' + \frac{|v_r|}{\lambda} \\ &= f_0' + \frac{|v_r|}{v/f_0'} \\ &= f_0' + f_0' \left(\frac{|v_r|}{v} \right) \\ &= f_0' \left(1 + \frac{|v_r|}{v} \right) \Rightarrow \end{aligned} \quad (4)$$

- (f) Die frekwensie van 'n ambulans se sirene is 1600 Hz. Die ambulans ry teen 95 km/h na die voet van 'n loodregte krans. Die spoed van klank in lug is 330 m/s.

The siren of an ambulance has a frequency of 1600 Hz. The ambulance is driving towards the base of a perpendicular cliff at 95 km/h. The speed of sound in air is 330 m/s.

- (i) Toon aan dat die frekwensie wat die pasiënt, wat van die krans afgeval het, sal hoor 1739 Hz sal wees wanneer die ambulans teen 95 km/h nader kom.

Show that the frequency that the patient, who had fallen from the cliff, will hear if the ambulance is approaching at 95 km/h will be 1739 Hz.

$$\begin{aligned} f' &= f_0 \left(\frac{1}{1 - v_s/v} \right) \\ 95 \text{ km/h} &= 95 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \div 60 \times 60 = 26,4 \text{ m/s} \\ \therefore f' &= 1600 \left[\frac{1}{1 - \frac{26,4}{330}} \right] = 1739 \text{ Hz} \Rightarrow \end{aligned} \quad (4)$$

- (ii) Bepaal die frekwensie f_1' van die waarkaatste golf wat die ambulansman sal hoor.

Determine the frequency f_1' of the reflected sound wave at the ambulance.

$$\begin{aligned} f_1' &= f' \left(1 + \frac{|v_r|}{v} \right) \\ &= 1739 \left[1 + \frac{26,4}{330} \right] \\ &= 1878 \text{ Hz} \end{aligned}$$

VRAAG/QUESTION 5

Die swaartepunt vir 'n aantal deeltjies, elk met massa Δm_i word gegee deur
The center of mass for a number of particles, each with mass Δm_i , is given by

$$x_{cm} = \frac{1}{M} \sum x_i \Delta m_i$$

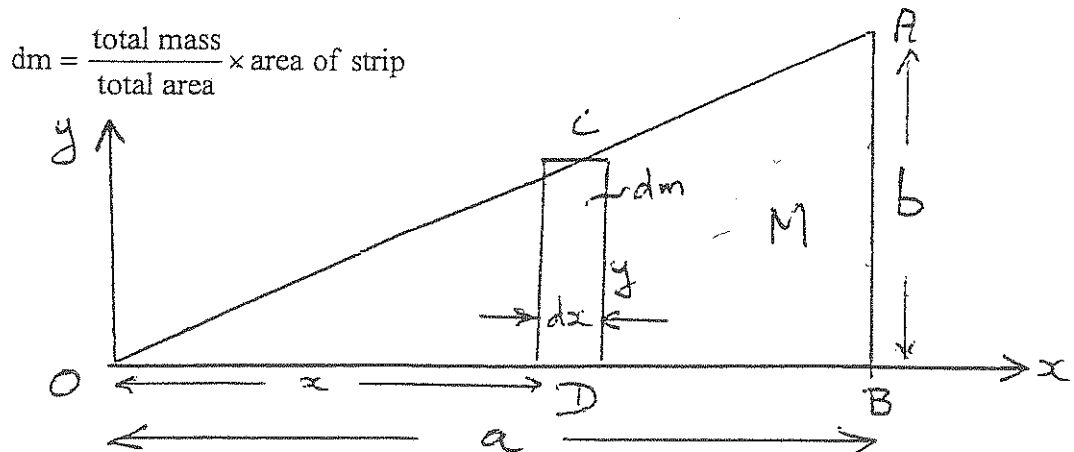
Vir 'n groot aantal deeltjies word hierdie vergelyking
For a large number of particles this equation becomes

$$x_{cm} = \frac{1}{M} \int x \, dm$$

Die volgende reghoekige driehoek se totale massa is M . Die massa dm van elke strokie word gegee deur.

$$dm = \frac{\text{totale massa}}{\text{totale oppervlakte}} \times \text{oppervlakte van strokie}$$

The following right triangular object has a total mass M . The mass dm of each strip is given by



- (i) Wat is die totale oppervlakte van ΔAOB ?
What is the total area of ΔAOB ?

$$\text{Area} = \frac{1}{2} ab$$

(1)

- (ii) Wat is die oppervlakte van die strokie met massa dm ?
What is the area of the strip with mass dm ?

$$\text{Strip Area} = y \, dx$$

(1)

- (iii) Skryf die massa dm in terme van:
Write the mass dm in terms of:

$M, y, dx, a, b.$

$$dm = \frac{M}{\frac{1}{2}ab} (y \, dx)$$

(2)

- (iv) Herskryf bogenoemde uitdrukking vir die x-koördinaat x_{cm} deur dm met die uitdrukking hierbo te vervang.

For the x coordinate, rewrite the expression for x_{cm} by replacing dm with the expression above.

$$\begin{aligned} x_{cm} &= \frac{1}{M} \int x \left(\frac{2M}{ab} y \, dx \right) \\ &= \frac{2}{ab} \int xy \, dx \end{aligned}$$

(2)

- (v) Uit die driehoek AOB en COD is:

From the similar triangles AOB and COD:

$$\frac{y}{x} = \frac{b}{a}, \text{ of / or } y = \frac{b}{a} x$$

Met die substitusie van y bepaal die waarde van die x-koördinaat van x_{cm} indien

$$\int x^2 dx = \frac{1}{3} x^3.$$

With this substitution, now determine the value of the x coordinate of x_{cm} if

$$\int x^2 dx = \frac{1}{3} x^3$$

$$\begin{aligned} x_{cm} &= \frac{2}{ab} \int x \left(\frac{b}{a} x \right) dx \\ &= \frac{2}{a^2} \int_0^a x^2 dx \\ &= \frac{2}{a^2} \left[\frac{1}{3} x^3 \right]_0^a \\ &= \frac{2}{a^2} \left(\frac{1}{3} a^3 \right) \\ &= \frac{2}{3} a \Rightarrow \end{aligned}$$

(3)

- (vi) Verwag u dat die y-koördinaat van die swaartepunt sal lê by:

Do you expect the y coordinate of the center of mass to be at

$$y_{cm} = \frac{2}{3} b \quad \square, \quad \frac{1}{3} b \quad \boxed{\times}, \quad \frac{1}{2} b \quad \square$$

(1)

[10]

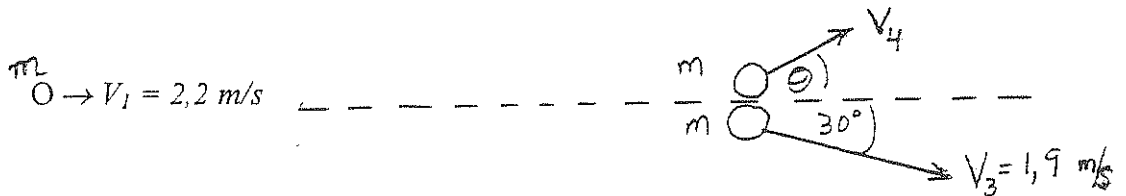
VRAAG/QUESTION 6

- (a) Wat is die voorwaarde(s) vir 'n elastiese botsing.
When is a collision known as elastic?

Momentum conserved (behoue)
KE " (") (2)

- (b) 'n Bal met massa $m = 0,5 \text{ kg}$ beweeg met 'n spoed $V_1 = 2,2 \text{ m/s}$ en bots teen 'n stilstaande bal met dieselfde massa m . Na die botsing beweeg een bal met 'n spoed $V_3 = 1,9 \text{ m/s}$ teen 'n hoek van 30° ten opsigte van die invalrigting. Die ander bal beweeg teen $V_4 \text{ m/s}$ met 'n hoek θ .

A ball with mass $m = 0,5 \text{ kg}$ is moving with a speed $V_1 = 2,2 \text{ m/s}$. It collides with a stationary ball with the same mass m . After the collision one ball moves at $V_3 = 1,9 \text{ m/s}$ in a direction at an angle of 30° with respect to the original line of motion. The other ball is moving at $V_4 \text{ m/s}$ at an angle of θ .



- (i) Bepaal θ en V_4 / Determine θ and V_4

X-rigting / X-direction $m_1 v_1 + m_2 v_2 = m_1 v_3 \cos 30^\circ + m_2 v_4 \cos \theta$

$$0,5(2,2) + 0 = 0,5(1,9) \cos 30^\circ + 0,5 v_4 \cos \theta$$

$$1,1 = 0,8227 + 0,5 v_4 \cos \theta$$

$$v_4 \cos \theta = 0,55 \Rightarrow$$

Y-rigting / Y-direction $m v_1 + m v_2 = m v_4 \sin \theta - m v_3 \sin 30^\circ$

$$0 + 0 = 0,5 v_4 \sin \theta - 0,5(1,9) \sin 30^\circ$$

$$v_4 \sin \theta = \frac{0,5(1,9) \sin 30^\circ}{0,5} = 0,95 \Rightarrow$$

$$\therefore \frac{v_4 \sin \theta}{v_4 \cos \theta} = \frac{0,95}{0,55}$$

$$\tan \theta = 1,71 \Rightarrow \theta = 60^\circ$$

$$v_4 \sin 60^\circ = 0,95 \quad ; \quad v_4 = 1,1 \text{ m/s} \Rightarrow \quad (9)$$

- (i) Bepaal of hierdie 'n elastiese of nie-elastiese botsing was.

Determine if this collision was elastic or inelastic.

$KE_{\text{before}} = \frac{1}{2} m v_1^2$ $= \frac{1}{2} (0,5) (2,2)^2$ $= 1,21 \text{ J} \Rightarrow$	$KE_{\text{after}} = \frac{1}{2} m v_3^2 + \frac{1}{2} m v_4^2$ $= \frac{1}{2} (0,5) (1,9)^2 + \frac{1}{2} (0,5) (1,1)^2$ $= 1,21 \text{ J} \Rightarrow$
$\therefore$ Elastic ✓ (4)	

- (ii) Watter % is die spoed V_4 relatief tot V_1 ?

Determine how much % is the speed V_4 relative to V_1

$$v_4/v_1 = \frac{1,1}{2,2} \times 100\% = 50\% \checkmark$$

(1)

- (iii) Watter % is die energie van V_4 relatief tot dié van V_1 ?

Determine the % of the energy of V_4 relative to that of V_1 .

$$\frac{\frac{1}{2} m v_4^2}{\frac{1}{2} m v_1^2} = \frac{\frac{1}{2} (0,5) (1,1)^2}{\frac{1}{2} (0,5) (2,2)^2} \times 100\% = 25\%$$

(1)

[17]